

# Efficiency and acceleration in the accumulation of children’s early vocabulary

Michael C. Frank

## Abstract

Children typically learn hundreds of words over the first few years of their life, in a process that begins slowly but quickly picks up speed. Yet children vary widely in their rate of word learning, and this variation has important practical and theoretical consequences. Prior models of word learning describe vocabulary growth as a process of accumulation of evidence over time. Here we show that the process is best characterized as *accelerating* accumulation, with children varying in both their overall learning efficiency (intercept) and their acceleration rate (slope). This model provides a lever for understanding different loci of variation. Applied to data from 24 languages, it suggests that the female advantage in early vocabulary results from differences in overall learning efficiency, while socioeconomic differences may compound through effects on acceleration rate. Using dense observations of children’s early language environment and word recognition speed, the model also suggests that differences in processing speed load on efficiency and do not compound, while differences in home language show some compounding effects on acceleration. In contrast to children, large language models – even those trained on child-directed speech – are not well-described by this model: they show standard accumulator dynamics with no acceleration and very limited variability. In sum, our model provides a quantitative characterization of key signatures of early vocabulary growth, providing a target for future mechanistic learning models.

Children’s early vocabulary grows very rapidly during early childhood, with large variation between children (Fenson et al. 1994, 2007; Frank et al. 2021). The growth has theoretical, practical, and methodological importance. From a theoretical perspective, growth mechanisms shed light on children’s fundamental learning mechanisms and how they change across childhood (McMurray 2007; Bloom 2002; Frank et al. 2021). From a practical perspective, early vocabulary is an important precursor – and predictor – of academic achievement (Marchman and Fernald 2008; Catts et al. 2002; Muter et al. 2004). From a methodological perspective, vocabulary also affords a unique psychometric opportunity to study human variation because, unlike nearly all important human psychological metrics, there is a true zero point – a time before which children know no words. Thus, absolute levels of variability can be calculated.

One dominant viewpoint is that children’s vocabulary growth can be characterized as a process of accumulation (McMurray 2007; Mitchell and McMurray 2009; Hidaka 2013; Mollica and Piantadosi 2017; Kachergis, Marchman, and Frank 2022). Each distinct word (type) can be conceptualized as a “bucket,” and every token of language input then is a “drop” that falls into its respective bucket. When each bucket is full, the word is learned (McMurray 2007; Kachergis, Marchman, and Frank 2022). These models have been used to recover interpretable differences in “bucket size,” often based on word properties including concreteness and syntactic category (Hidaka 2013; Kachergis, Marchman, and Frank 2022). They also provide a graded explanation for the widely-observed “vocabulary spurt” phenomenon – in which children’s rate of vocabulary learning appears to increase

dramatically (Frank et al. 2021; Ganger and Brent 2004; Gómez Díaz et al. 2024).

The accumulator view connects with a viewpoint in which the quantity of language that children hear plays a strong causal role in the speed of their learning (Fernald and Marchman 2012). Indeed, there is systematic evidence that variation in children’s input – both its *quantity* and its *quality* – is related to variation in vocabulary (Hart and Risley 1995; Fernald and Marchman 2012; Weisleder and Fernald 2013; Anderson et al. 2021; Rowe 2012; Hirsh-Pasek et al. 2015; Coffey and Snedeker 2026). While input quantity is relatively easy to measure, measuring what makes input high quality has been challenging. One influential framework suggests that high quality input is interactionally rich, linguistically appropriate for the level of the child, and conceptually complex (Rowe and Snow 2020).

A naïve interpretation of the accumulator view would suggest that substantial variance in vocabulary growth should be related to input – either via quantity (number of drops) or quality (size of each drop). Two recent meta-analyses complicate this view, however. Anderson et al. (2021) showed that only a modest proportion of the variance in vocabulary outcomes is related to input quantity ( $r^2 = .11$ ) and quality ( $r^2 = .04$ ). A newer meta-analysis of quantity confirms and extends the proportion of variance due to quantity to 4–7% of total variation in vocabulary (Coffey and Snedeker 2026). Thus – as we might have supposed based on other research on human variation – not all variation in vocabulary between children can be attributed to input differences.

Further, work on children’s word recognition suggests that children become faster and more accurate at matching words to their referent across early childhood (Bergelson 2020; Fernald et al. 1998; Fernald, Perfors, and Marchman 2006; Peter et al. 2019; Frank et al. 2026). Much of this work relies on the “looking while listening” (LWL) paradigm, in which children see two pictures (e.g. a book and a car) and their eye movements are recorded as they hear a sentence that refers to one of the pictures (e.g., “look at the book”). Children’s reaction time to shift their eyes to the target (the book) provides an index of their speed of word recognition that decreases logarithmically across early childhood (Fernald et al. 2008; Frank et al. 2026) and relates to their vocabulary and other language outcomes (Fernald, Perfors, and Marchman 2006; Marchman and Fernald 2008; Fernald and Marchman 2012; Fernald, Marchman, and Weisleder 2013; Peter et al. 2019; Frank et al. 2026). One hypothesis proposed in this work is that greater language input leads to more rapid processing, in a virtuous cycle in which faster processing in turn leads to greater word learning (Weisleder and Fernald 2013). In the accumulator metaphor, one way to describe this proposal would be that individual children’s buckets can be bigger or smaller based on their prior input and their processing efficiency, but to our knowledge this suggestion has not been fleshed out.

As this discussion indicates, despite the promise of accumulator models, they have largely not been used to understand variation between individuals. Instead, the majority of applications have focused on variation between words (Hidaka 2013; Kachergis, Marchman, and Frank 2022). Further, the rapidity of children’s early word learning stands as a rebuke to a naïve accumulator view. Sometimes exposure to a small number of examples of a word in the lab is enough for a child to form a lasting (Markson and Bloom 1997; Woodward, Markman, and Fitzsimmons 1994). In some cases, it seems only a small number of high-quality instances is sufficient for learning (Trueswell et al. 2013; Mollica and Piantadosi 2017), but if so, why do some words take much longer to learn?

The goal of the current work is to address these puzzles through systematic investigation and extension of accumulator models. All accumulator models seek to explain the same pattern: that

the overall size of children’s vocabulary increases supra-linearly. Where they differ is in what this growth means for the underlying learning process. Following previous work, we start with the idea that children’s *internal learning ability* (not just their observed supra-linear vocabulary growth) accelerates across development (Hidaka 2013; Mayor and Plunkett 2010). We then extend a prior psychometric accumulator model (Kachergis, Marchman, and Frank 2022): our key addition is to describe variation between individual children by modeling both their individual learning efficiency and the acceleration of their learning – following the bucket metaphor and allowing children to have different size buckets. This extension defines a longitudinal growth model that can be fit directly to large-scale datasets about children’s actual vocabularies.

To constrain our model, we use data from the MacArthur-Bates Communicative Development Inventories [MB-CDI; Fenson et al. (1994); Marchman, Dale, and Fenson (2021)], a popular parent report instrument in which parents are asked to indicate which words their child produces and understands. Here we focus exclusively on production, which is used with older children and which tends to be a more reliable measure (Frank et al. 2021). We make use of longitudinal data from Wordbank (Frank et al. 2017), an open repository of data from the MB-CDI.

To address questions about the relationship between language input, language processing, and growth, we fit a Bayesian version of our growth model. We make use of new data resources to estimate the magnitude of linkages between input and vocabulary growth. First, we estimate plausible population values based on independent estimates of population variation in language input (Coffey et al. 2024). In addition, we use measures of individual children’s language input from egocentric videos (Long et al. 2024) and day-long home recordings (Egan-Dailey and Bergelson 2025; Adams et al. 2018; Fernald, Marchman, and Weisleder 2013) to estimate the input-uptake relationship more precisely for a subsample of children, and measures of LWL word recognition from Peekbank (Zettersten et al. 2023), a large scale repository of LWL data.

Overall, our analysis supports a view in which children’s vocabulary growth is not a pure process of accumulation but rather one that accelerates rapidly over time. We end by comparing the fit of our models on human data to their fit on large-language models. Making use of GPT-2 models trained on human developmental data, we show that – in contrast to children – these models *do* appear to function as pure accumulators. This analysis exposes a critical disanalogy between LLMs and human learners and suggests a potential route for understanding the vast “data gap” between children and large language models (Frank 2023; Warstadt et al. 2023).

## Model

To begin our analysis, we present a nested set of accumulator-type models, many of which have been presented in the literature (see SI: Comparison with Prior Models). As we show below, each refinement of the model contributes to its ability to describe the variability between children. We start with the Rasch model from psychometrics (Rasch 1960; Bond and Fox 2013), which assumes that each child  $i$  has a learning ability  $\theta_i$  and each word  $j$  has a difficulty  $\delta_j$ .<sup>1</sup> The probability that

---

<sup>1</sup>A variety of previous work has focused on estimating the properties of individual words (Goodman, Dale, and Li 2008; Hidaka 2013; Roy et al. 2015; Braginsky et al. 2019; Kachergis, Marchman, and Frank 2022; Kachergis et al. 2022). These analyses reveal that there is predictable word-level difficulty information but that only a modest portion of the variation between words is carried by frequency information. Words can be hard for a child to acquire because they vary also in the complexity and concreteness of their meanings, their syntactic category and syntactic context,

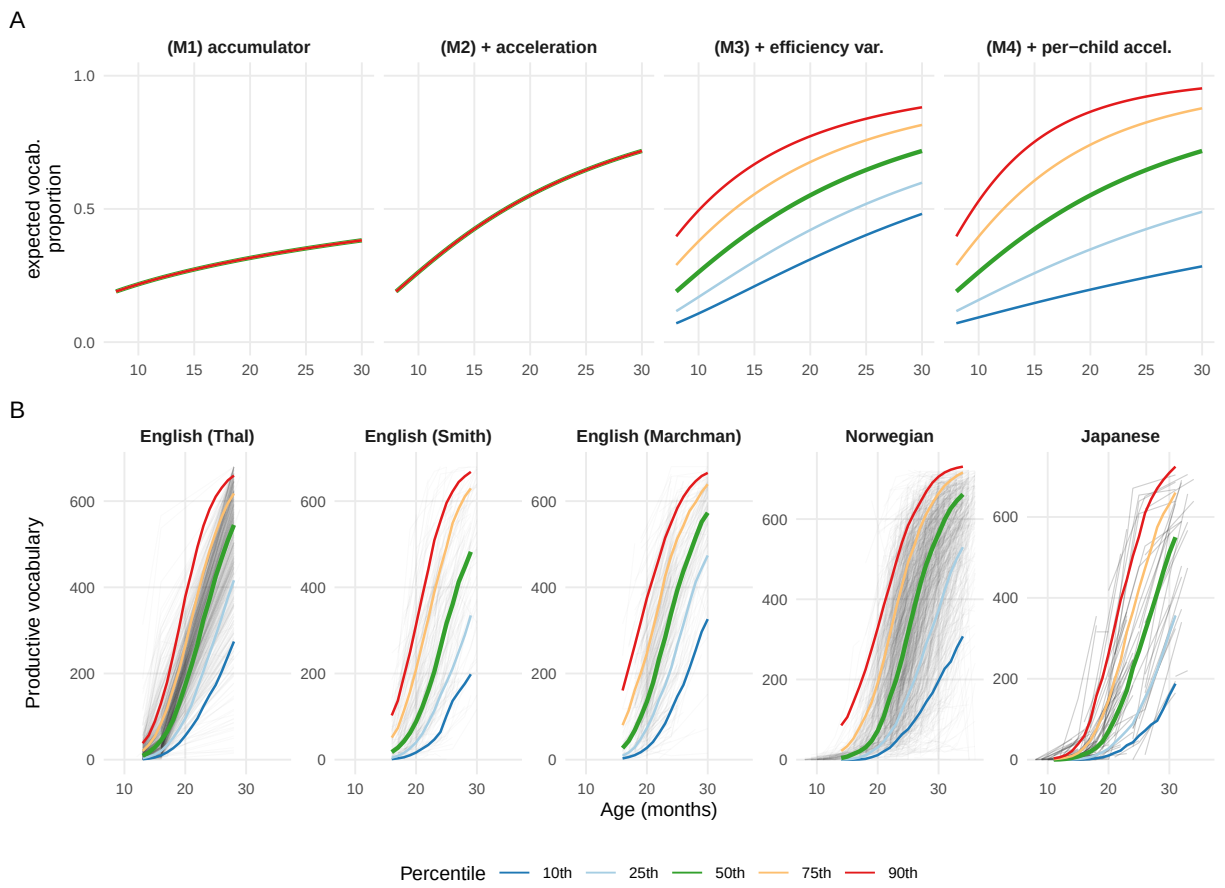


Figure 1

child  $i$  produces word  $j$  is

$$P(w_{i,j} = 1 \mid \theta_i, \delta_j) = \frac{\exp(\theta_i - \delta_j)}{1 + \exp(\theta_i - \delta_j)}$$

Following Kachergis, Marchman, and Frank (2022), we can write our first model as a Rasch-style accumulator model (Model 1), specifying that  $\theta_i$  is not a single parameter but instead a function of accumulation over time. In the simplest form:

$$\text{M1: } \theta_i(t) = \log(r_i) + \log(t/a_0)$$

where  $r_i$  is the child’s rate of language input (in tokens per month) and  $t$  is the child’s age (in months, anchored at  $a_0$ ); note that we use log age to simulate linear accumulation because  $\exp(\theta) = r \cdot t$ . This form instantiates the basic hypothesis that the only thing that changes over time is more input, and the only source of variation between children is variation in rate.

We compare this model to an *accelerating accumulator* model (Model 2), in which the amount learned from each exposure increases with time by an exponent  $\kappa$ .

$$\text{M2: } \theta_i(t) = \log(r_i) + \kappa \log(t/a_0)$$

When  $\kappa > 1$ , each successive month yields more accumulation than the last.

Neither of these models has any way to capture variation across children beyond variation in input rate. Thus, we augment the basic accelerating accumulator, rewriting it as the sum of an individually variable efficiency term:

$$\text{M3: } \theta_i(t) = \xi_i + \kappa \log(t/a_0)$$

The intercept term  $\xi_i = \log(r_i) + \log(\alpha_i)$  can then be used to break down individual variation into both input rate-related and other variation in children’s efficiency.

Our final model includes individually variable efficiency and acceleration terms:

$$\text{M4: } \theta_i(t) = \xi_i + \kappa_i \log(t/a_0)$$

In this model, the slope term  $\kappa_i = \kappa + \gamma \log r_i + \zeta_i$  adds *individual variation in acceleration*, decomposing acceleration into a population term, input-related acceleration, and a child-specific residual acceleration  $\zeta_i$  (the part not explained by input). The coefficient  $\gamma$  then captures the strength of the relationship between input and acceleration.

Model 4 instantiates the idea that children vary in their base rate of efficiency and their acceleration, creating variant of a standard longitudinal growth model in which children vary on their slope and intercept (Grimm, Ram, and Estabrook 2016). Note that, while we define the relationship of  $\xi_i$  and

---

their phonology, and many other factors. In the current work, we estimate word difficulty as a free parameter, and focus instead on variation between children.

$\kappa_i$  to a child’s input  $r_i$ , we can nevertheless fit the simplest form of the model by just allowing these parameters to vary as random child-level slopes and intercepts.

As we will show, Model 4 provides a good description of variation in children’s longitudinal growth trajectories. Figure 1 (top) shows schematic predictions from a pure accumulator model (M1), an accelerating accumulator (M2) in comparison to models including child-level variation in efficiency (M3) and both efficiency and acceleration (M4). Each model’s predictions are represented as ability scores ( $\theta$ ) across age and then also projected into proportion correct scores on a discrete set of vocabulary items, representing an observed accuracy score that is subject to compression at floor and ceiling.

Our analysis generally depends on the availability of longitudinal data at the item level. The different models described above are not distinguishable from the cross-sectional, aggregate data that are typically considered in many analyses of early vocabulary. First, aggregate data do not allow for the estimation of individual word difficulties, thus the distribution of difficulties remains latent and different difficulty distributions can be posited to explain the accelerating shape of vocabulary growth (McMurray 2007; Mitchell and McMurray 2009). Second, cross-sectional data do not allow for identification of whether individual children’s variation is due to differences in efficiency (intercept) or acceleration (slope) (see SI: Nonidentifiability in Cross-Sectional Data).

## Results

### Children’s longitudinal growth reflects differences in both efficiency and acceleration

We first retrieved longitudinal data on monolingual, typically-developing children’s vocabulary from Wordbank (Frank et al. 2017), an open archive of data from the MacArthur-Bates Communicative Development Inventory and its international adaptations. SI: Datasets shows the datasets used in each study in the paper; in this first analyses, we made use of five longitudinal datasets from Wordbank (three from American English and one each from Norwegian and Japanese,  $N_{total} = 3135$ ).

Taking advantage of the equivalence of Rasch models to generalized linear mixed effects models (De Boeck et al. 2011), we fit the set of models shown in Figure 1 as a series of multi-level logistic regressions. Each model predicted the probability of an individual child producing a particular word as a function of age (either unmodified in M1 or multiplied by a constant in M2) and a random effect of word difficulty. Successive models add random intercepts (M3) and slopes (M4) by child. This initial set of fits simply attempts to describe the distribution of vocabulary growth curves.

Model 4, which incorporated variation in acceleration and efficiency, was the best fitting model across all datasets with differences in AIC  $> 1000$  in all cases (see SI: AIC and BIC for Longitudinal Models). Figure 1 (bottom) shows the fit of the four model variants; the increasing spread of children’s trajectories with age is especially noticeable for low-vocabulary older children [who may be classified as “late talkers”; Fernald and Marchman (2012); Rescorla (2011)].<sup>2</sup>

---

<sup>2</sup>Following the literature, we primarily investigate linear accumulator models, which show power-law growth of ability over time because  $\theta$  is a function of  $\log(t)$ . However, it is also possible that ability grows linearly in time, which would lead to exponential increases in the probability of word production. We fit linear-time models, but found that they were dispreferred statistically in all but one case, though differences were relatively small within the restricted age range for which we have data (See SI: Full Cross-linguistic Model Ladder).

## Efficiency and acceleration relate to separate demographic predictors

Children’s vocabulary is known to be related to a number of sociodemographic predictors, including sex and maternal education (which is interpreted as a proxy for socioeconomic status). In particular, female children show consistently larger vocabularies than male children across dozens of languages and countries (Eriksson et al. 2012; Frank et al. 2021). Similarly, children of more educated mothers tend to show larger vocabularies, though this effect is somewhat more heterogeneous across countries and languages (Frank et al. 2021).

Our model offers a summary of children’s growth in terms of both their efficiency (intercept) and their acceleration (slope). This decomposition allows us to ask whether these two demographics effects – sex and maternal education – are seen in efficiency, acceleration, or both. For each child in our analysis, we extracted the best linear unbiased prediction (BLUP) from our models. We then regressed intercept and slope terms against each demographic predictor separately for each language. Although we cannot fit individual children’s growth trajectories in cross-sectional data, we can also in principle apply the same model to recover population parameters based on cross-sectional data, so we did so for an additional group of datasets for which sex and maternal education information were available.

Figure 2 shows the per-language coefficients from this analysis, along with a cross-language meta-analytic estimate for both the cross-sectional and longitudinal data (which largely agree). Examining the cross-sectional data since this analysis includes more datasets, the female advantage is apparent across all datasets, but this effect manifests itself more substantially efficiency ( $\beta = 0.54$ ,  $p = < .001$ ) than acceleration ( $\beta = 0.35$ ,  $p = < .001$ ). In contrast, maternal education effects were present in both measures, but stronger in acceleration ( efficiency:  $\beta = 0.1$ ,  $p = .004$ ; acceleration:  $\beta = 0.38$ ,  $p = < .001$ ). One interpretation of these findings is that sex effects reflect a small constant offset in average ability. In contrast, socioeconomic differences – as reflected in maternal education – might continue to compound across development (Fernald, Marchman, and Weisleder 2013; Hart and Risley 2003).

## Population input-related variation corresponds to meta-analytic estimates

In our prior model fits, we fit population variation in efficiency and acceleration without considering the connection to language input. We next consider whether this relationship can be estimated at a population level. We use markov chain monte carlo to fit a version of model 4 with linkages to input rate.

We first estimate these linkages at the population level by assuming a known level of population variation in language input rate:  $\sigma_r$ . We can then examine the variance due to input variation vs. population by fitting the parameter  $\pi_\alpha = \frac{\sigma_\alpha^2}{\sigma_\alpha^2 + \sigma_r^2}$ , which defines the proportion of population variation in efficiency due to input variation. Importantly, this parameter is directly comparable to the meta-analytic estimates reported earlier (Anderson et al. 2021; Coffey and Snedeker 2026). This model – which does not include individual child-level input data – can not distinguish input contributions to efficiency vs. acceleration, but it provides a large dataset to begin analyzing the question.

Estimating children’s input range is notoriously tricky, since the rate of language that children hear varies across cultures, families, activities, and measurement occasions (Coffey et al. 2024). To

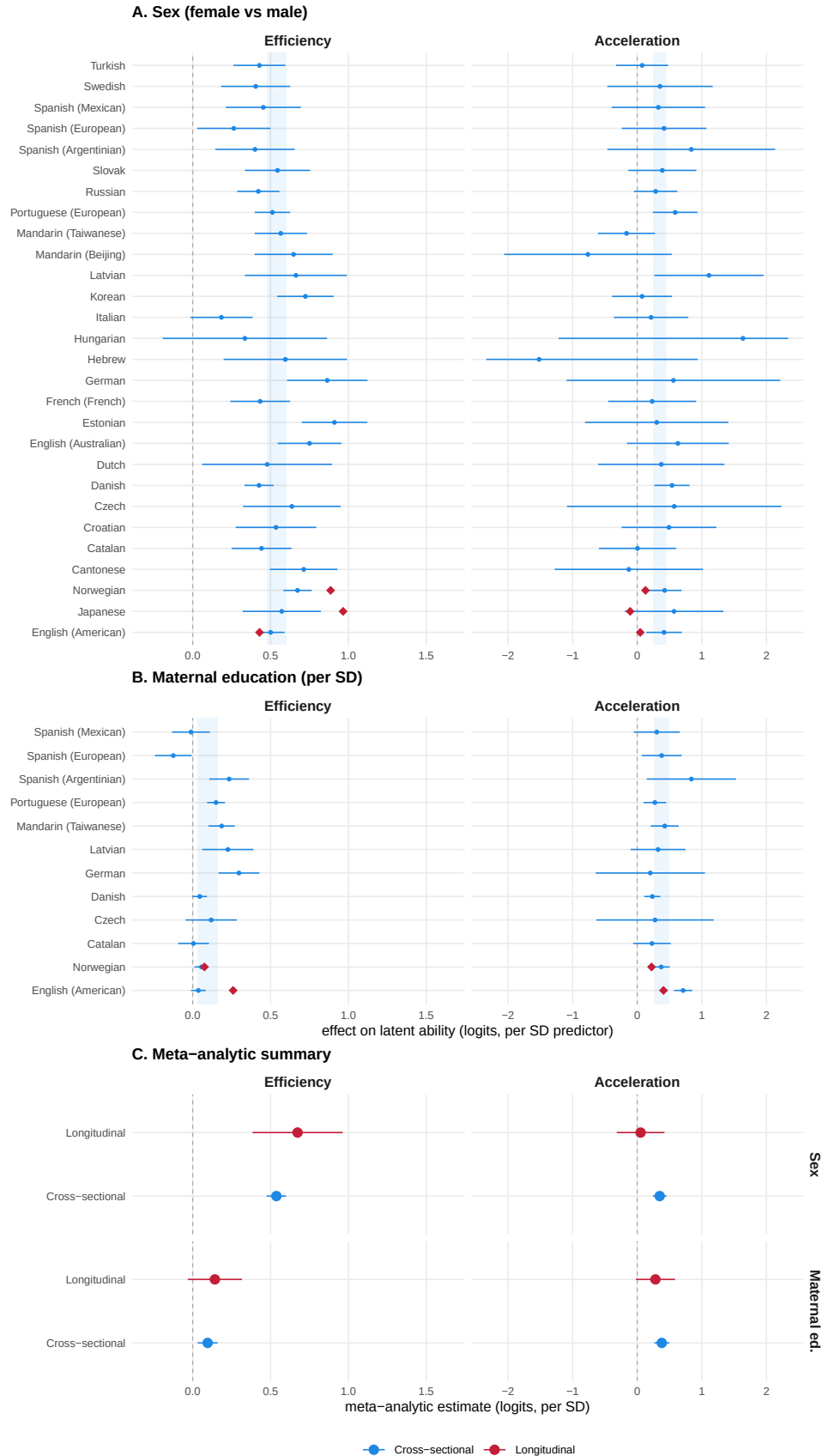


Figure 2: Regression coefficients predicting individual children's efficiency and acceleration as a function of gender and maternal education. Panels A and B show data from individual languages (both longitudinal and cross-sectional); blue regions show meta-analytic estimates from a random-effects meta analysis. Error bars show 95% confidence intervals. Panel C shows a meta-analytic summary for cross-sectional and longitudinal datasets



anchor our model using a range of population values for  $\sigma_r$ , we surveyed the literature for plausible estimates (see SI: Population Input Range Estimation). In our main simulations we set  $\sigma_r = .53$ , a value drawn from Sperry, Sperry, and Miller (2019), who estimated a mean of 419k tokens per month ( $\pm 1$  SD: 329k – 957k). We provide sensitivity analysis for alternative values based on an analytic computation of the relationship between  $\sigma_r$  and  $\pi_\alpha$  (see SI:  $\sigma_r$  Sensitivity Analysis).

Figure 3 shows the results of this analysis for both English and Norwegian longitudinal data (combined  $N=2957$ ).  $\pi_\alpha$  was 0.923 (CI: [0.917, 0.929]) for English and 0.961 (CI: [0.958, 0.964]) for Norwegian, both largely consistent with the previous meta-analyses. Sensitivity analysis showed that only implausibly large input variation would produce substantially higher values.

### Input predicts both efficiency and acceleration, while processing speed relates only to efficiency

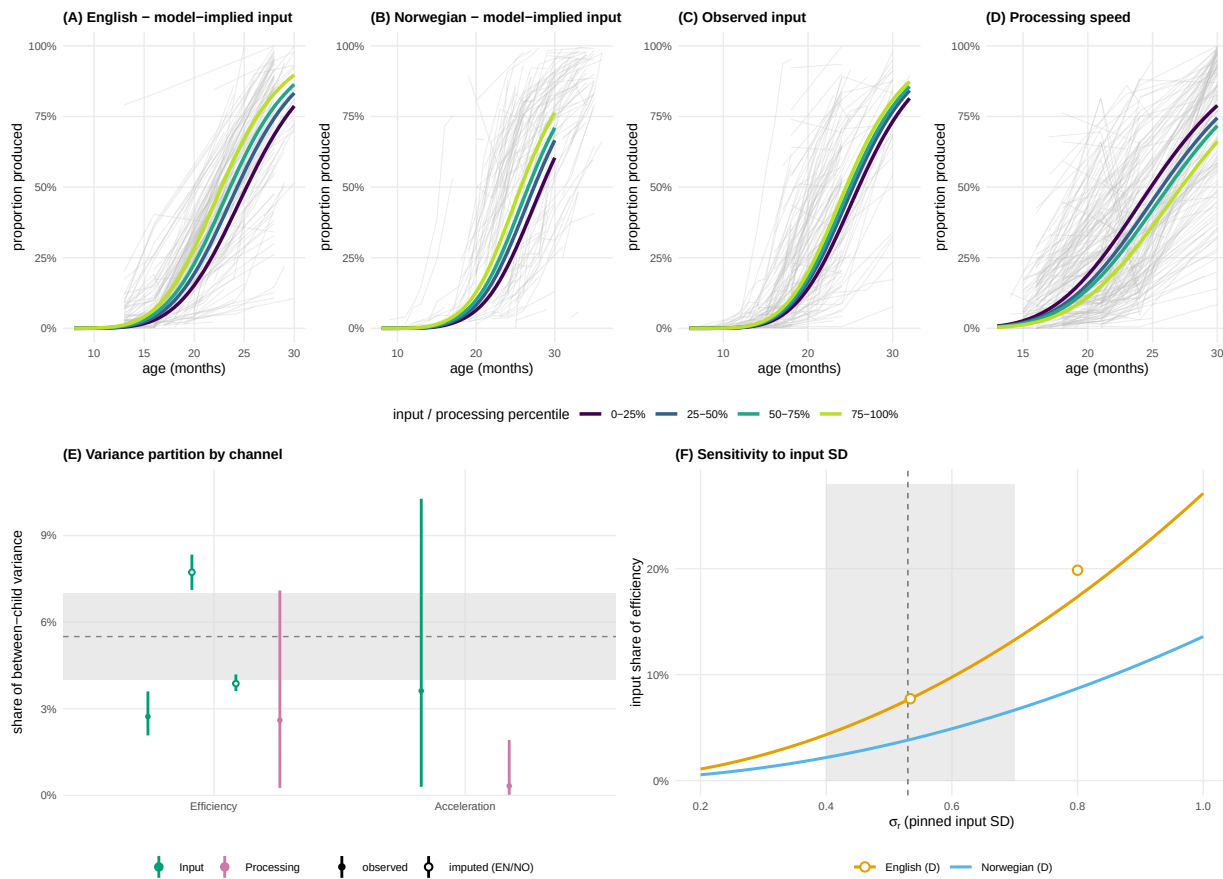


Figure 3

While the population estimates of input-sensitivity given above match our expectations from meta-analyses, they do not offer a direct estimate of input couplings within individual children, and they do not allow us to estimate the separate contributions of language input and language processing to efficiency and acceleration.

For this, we turn to a set of five longitudinal datasets that include either dense home recordings,

longitudinal LWL processing data, or both. Three include recordings from day-long audio recordings using LENA (Language Environment Analysis) recorders, which offer an automated adult word count estimate that has been validated by transcription studies (Cristia et al. 2021). One includes short video recordings of egocentric video recorded in children’s homes, with word counts estimated from automated transcripts (Long et al. 2024). Each of these metrics provides a noisy proxy for total home input, but nevertheless allows us to estimate models that include individual input. And four of the five include multiple LWL sessions per individual.

We thus fit the same model linking input estimates to variation in efficiency and acceleration, but this time with observations of  $r_i$ , for each child, allowing us to fit the full version of Model 4 described above, in which language input could contribute to both efficiency and acceleration. We additionally a

Figure 3 (panel C) shows a small separation of trajectories by input quartile, but the key question of interest is how this variation partitions across efficiency and acceleration.

Results for  $\pi_\alpha$  and  $\pi_\zeta$  are shown in Figure 3. Overall, we observe small but highly reliable effects of language input on both efficiency and acceleration. These effects are also in the range of the meta-analyses, but they suggest that input-related differences – though small – could compound through their effect on acceleration.

### Growth in ability leads to rapid changes in learning times for individual words

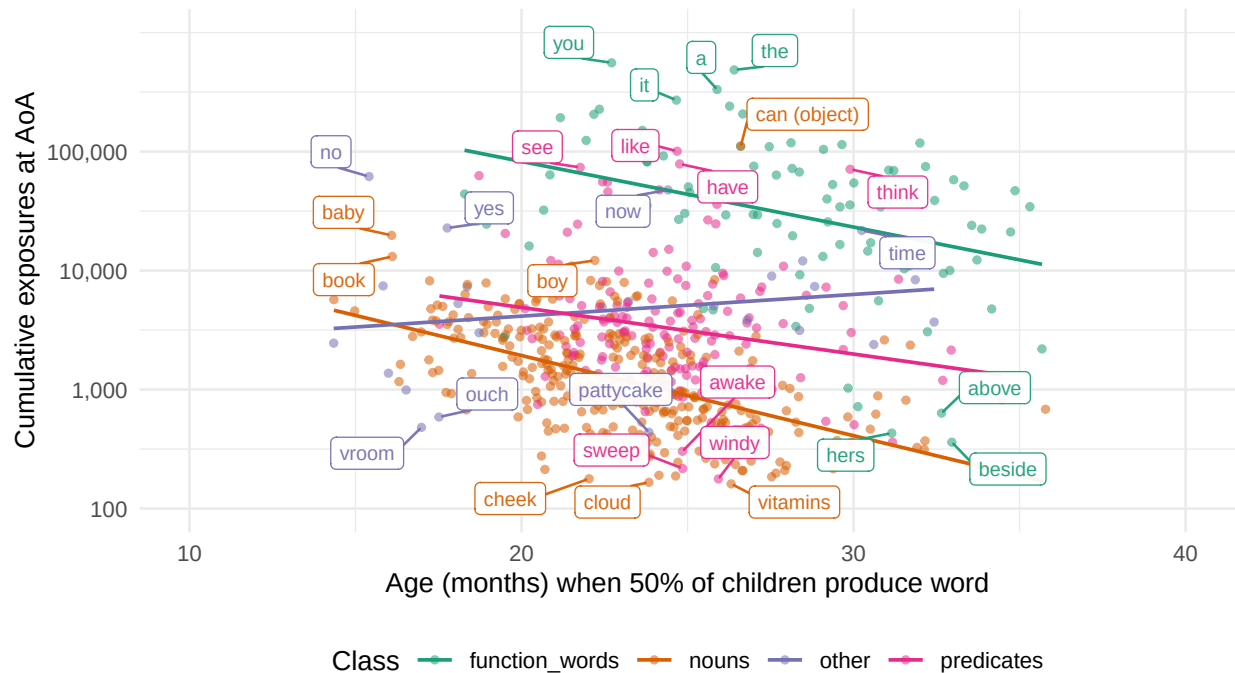


Figure 4

As an illustration of the changes in efficiency and acceleration that we observe in our data, we computed the average proportion of exposures necessary for a child to learn a word across ages. Figure 4 shows a model-derived estimate of the average number of times a word is heard before more

than 50% of children produce it (See SI: Changes in Word Efficiency), using frequency estimates from the CHILDES Corpus (MacWhinney 2000; Sanchez et al. 2019).

Across different word classes (e.g., nouns, verbs, closed class words), efficiency increases exponentially, such that early-learned words such as “book” may be heard tens of thousands of times before they are produced, while later learned words such as “vitamin” might only be heard hundreds times before being produced. Indeed, this analysis likely understates the degree of change in efficiency as it assumes that each exposure to a word is equally meaningful. In fact, individual exposures vary widely in how much information they give. Sometimes words are overheard rather than directed to the child, or lack physical context to allow for clear inference (Cartmill et al. 2013; Trueswell et al. 2013; Mollica and Piantadosi 2017). Thus, in line with both intuition and experiments, the total number of exposures required on average to learn a word might indeed be only a small handful by age three (Markson and Bloom 1997; Bloom 2002).

## Large language models do not show acceleration in their vocabulary growth

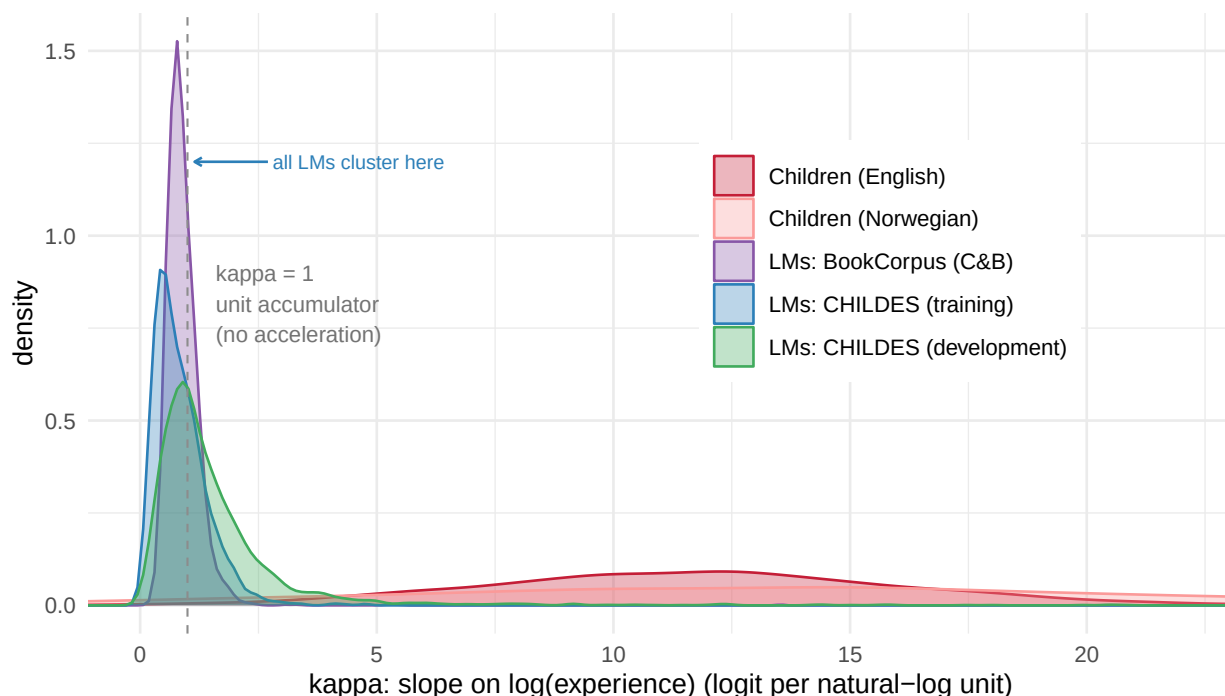


Figure 5: Acceleration slopes ( $\kappa$  = slope of competence on log experience). Children’s by-child  $\kappa_i$  (English, Norwegian) sit  $\sim 10\times$  above the  $\kappa = 1$  unit-accumulator boundary. Language models’ by-word slopes cluster at  $\kappa \sim 1$  across three populations: BookCorpus models from Chang & Bergen (2022, training axis), and our CHILDES-trained GPT-2 on both the training axis and the distinct-input development axis (converged-model ladder).

Small-scale language models trained on developmentally-realistic corpora have become a useful tool for understanding early language development (Huebner et al. 2021; Futrell and Mahowald 2025; Frank and Goodman 2025). For our final experiments, we were interested in whether large language models show the same patterns of developmental change in efficiency or acceleration. We made use of work by Chang and Bergen (2022), who extracted per-word acquisition curves from language

models. They computed logistic fits to the surprisal (negative log probability) of individual words across training, estimating

We can compare

Evanson, Lakretz, and King (2023) investigated

We end by providing a quantitative comparison between within-child human learning and LLM scaling laws, showing that children’s early language learning is strikingly different from LLM scaling: it shows both steeper scaling and greater variance in growth rate.

## Discussion

What we did

Longitudinal growth models efficiency and acceleration. Two different parameters.

Where does acceleration come from? Could come from a variety of sources, including general maturation of memory and attention, specific changes in word recognition or processing (Fernald et al. 1998; Frank et al. 2026), the increasing sophistication of inferential strategies for word learning [Smith et al. (2002); Markman and Wachtel (1988); etc], or other sources.

Mitchell and McMurray (2009) prove that acceleration does not result from simply leverage of vocabulary –

Role of input: Coffey 2026 variance decomposition

Furthermore, how does the accumulation process relate to the rapid growth of vocabulary? On some models, accumulation by itself can produce an exponential “vocabulary explosion” while the child’s own learning abilities remain constant. On other accounts, however, the child’s own abilities change over development. These changes could be due to the accumulation of language allowing bootstrapping (e.g., via mutual exclusivity or the shape bias), they could be through faster processing (Fernald rich get richer), or they could be via general maturation (e.g., better memory, faster processing). Or all of these.

Thus, modeling human learning requires positing other sources of variability and acceleration. Sources of variability could include from variation in interactional input quality, genetic variation between children, environmental richness not captured by input, etc. Sources of acceleration could include increases in processing efficiency, learning-to-learn including mutual exclusivity, shape bias, syntactic bootstrapping, as well as domain general changes in perception, processing speed, memory, attention or other aspects of cognition. The key to understanding children’s data efficiency may thus lie in better understanding when and how children’s learning accelerates with maturation and experience.

Continuous variation or clusters? (Pinstock and Schramm 2026; Singletary et al. 2025) We think it’s just continuous variation (Supplement)

Network models and their predictions e.g. (Steyvers and Tenenbaum 2005; Hills et al. 2009; Fourtassi, Bian, and Frank 2020)

# Methods

## Data and Code

Data were retrieved from Wordbank using the `wordbankr` package (Frank et al. 2017), childes-db VER using `childesr` (Sanchez et al. 2019), and peekbank version 2026.1 using `peekbankr` (Zettersten et al. 2023). BabyView data are from release 2025.2, specifically those used by (tan2026?). SEEDLings data are downloaded from the materials provided by Egan-Dailey and Bergelson (2025).

Reproducible code for this paper is available at <http://github.com/langcog/standard-model-2>.

## Models

### Frequentist models

Initial models were fit with `glmer` (Bates et al. 2015), with the following specifications:

- M1: `produces ~ offset(log_age) + (1 | item)`
- M2: `produces ~ 1 + log_age + (1 | item)`
- M3: `produces ~ 1 + log_age + (1 | item) + (1 | child)`
- M4: `produces ~ 1 + log_age + (1 | item) + (log_age | child)`

Linear variants reported in the supplement used linear age rather than log age.

### Bayesian models

Bayesian models were fit using Stan (Carpenter et al. 2017).

Priors

Inference

### Input

### GPT-2 models

We trained XYZ

## Acknowledgements

Some of the computing for this project was performed on the Sherlock and Marlowe clusters. We would like to thank Stanford University and Stanford Research Computing for providing computational resources and support that contributed to these research results. Funds for computational experiments were also provided by a gift from Meta to support research on children’s home language environment. Claude Code was used for literature review, code generation, visualization, and simulation infrastructure.

## References

- Adams, Katherine A., Virginia A. Marchman, Elizabeth C. Loi, Melanie D. Ashland, Anne Fernald, and Heidi M. Feldman. 2018. "Caregiver Talk and Medical Risk as Predictors of Language Outcomes in Full Term and Preterm Toddlers." *Child Development* 89 (5): 1674–90.
- Anderson, Nina J., Susan A. Graham, Heather Prime, Jennifer M. Jenkins, and Sheri Madigan. 2021. "Linking Quality and Quantity of Parental Linguistic Input to Child Language Skills: A Meta-Analysis." *Child Development* 92 (2): 484–501.
- Bates, Douglas, Martin Mächler, Ben Bolker, and Steve Walker. 2015. "Fitting Linear Mixed-Effects Models Using lme4." *Journal of Statistical Software* 67 (1): 1–48. <https://doi.org/10.18637/jss.v067.i01>.
- Bergelson, Erika. 2020. "The Comprehension Boost in Early Word Learning: Older Infants Are Better Learners." *Child Development Perspectives* 14 (3): 142–49.
- Bloom, Paul. 2002. *How Children Learn the Meanings of Words*. MIT press.
- Bond, Trevor G, and Christine M Fox. 2013. *Applying the Rasch Model: Fundamental Measurement in the Human Sciences*. Psychology Press.
- Braginsky, Mika, Daniel Yurovsky, Virginia A Marchman, and Michael C Frank. 2019. "Consistency and Variability in Children's Word Learning Across Languages." *Open Mind* 3: 52–67.
- Carpenter, Bob, Andrew Gelman, Matthew D. Hoffman, Daniel Lee, Ben Goodrich, Michael Betancourt, Marcus A. Brubaker, Jiqiang Guo, Peter Li, and Allen Riddell. 2017. "Stan: A Probabilistic Programming Language." *Journal of Statistical Software* 76 (1): 1–32. <https://doi.org/10.18637/jss.v076.i01>.
- Cartmill, Erica A, Benjamin F Armstrong III, Lila R Gleitman, Susan Goldin-Meadow, Tamara N Medina, and John C Trueswell. 2013. "Quality of Early Parent Input Predicts Child Vocabulary 3 Years Later." *Proceedings of the National Academy of Sciences* 110 (28): 11278–83.
- Catts, Hugh W, Marc E Fey, J Bruce Tomblin, and Xuyang Zhang. 2002. "A Longitudinal Investigation of Reading Outcomes in Children with Language Impairments." *Journal of Speech, Language, and Hearing Research* 45 (6): 1142–57.
- Chang, Tyler A., and Benjamin K. Bergen. 2022. "Word Acquisition in Neural Language Models." *Transactions of the Association for Computational Linguistics* 10: 1–16. [https://doi.org/10.1162/tacl\\_a\\_00444](https://doi.org/10.1162/tacl_a_00444).
- Coffey, Joseph R., Okko Räsänen, Camila Scaff, and Alejandrina Cristia. 2024. "The Difficulty and Importance of Estimating the Lower and Upper Bounds of Infant Speech Exposure." In *Proceedings of Interspeech 2024*, 3615. Kos, Greece. <https://doi.org/10.21437/Interspeech.2024-2102>.
- Coffey, Joseph R., and Jesse Snedeker. 2026. "How Strong Is the Relationship Between Caregiver Speech and Language Development? A Meta-Analysis." *Journal of Child Language* 53: 435–70. <https://doi.org/10.1017/S0305000924000692>.
- Cristia, Alejandrina, Marvin Lavechin, Camila Scaff, Melanie Soderstrom, Caroline Rowland, Okko Räsänen, John Bunce, and Erika Bergelson. 2021. "A Thorough Evaluation of the Language Environment Analysis (LENA) System." *Behavior Research Methods* 53 (2): 467–86.
- De Boeck, Paul, Marjan Bakker, Robert Zwitser, Michel Nivard, Abe Hofman, Francis Tuerlinckx, and Ivailo Partchev. 2011. "The Estimation of Item Response Models with the Lmer Function from the Lme4 Package in r." *Journal of Statistical Software* 39: 1–28.
- Egan-Dailey, Shannon, and Erika Bergelson. 2025. "Early Child Measures Outpredict Input Measures of Preschool Language Skills in US English Learners." *Developmental Psychology*.

- Eriksson, Mårten, Peter B Marschik, Tiia Tulviste, Margareta Almgren, Miguel Pérez Pereira, Sonja Wehberg, Ljubica Marjanović-Umek, Frederique Gayraud, Melita Kovacevic, and Carlos Gallego. 2012. “Differences Between Girls and Boys in Emerging Language Skills: Evidence from 10 Language Communities.” *British Journal of Developmental Psychology* 30 (2): 326–43.
- Evanson, Linnea, Yair Lakretz, and Jean-Rémi King. 2023. “Language Acquisition: Do Children and Language Models Follow Similar Learning Stages?” In *Findings of the Association for Computational Linguistics: ACL 2023*, 12205–18. Toronto, Canada: Association for Computational Linguistics. <https://doi.org/10.18653/v1/2023.findings-acl.773>.
- Fenson, Larry et al. 2007. *MacArthur-Bates Communicative Development Inventories*. Baltimore, MD: Paul H. Brookes Publishing Company.
- Fenson, Larry, Philip S Dale, J Steven Reznick, Elizabeth Bates, Donna J Thal, Stephen J Pethick, Michael Tomasello, Carolyn B Mervis, and Joan Stiles. 1994. “Variability in Early Communicative Development.” *Monographs of the Society for Research in Child Development*, i–185.
- Fernald, Anne, and Virginia A Marchman. 2012. “Individual Differences in Lexical Processing at 18 Months Predict Vocabulary Growth in Typically Developing and Late-Talking Toddlers.” *Child Development* 83 (1): 203–22.
- Fernald, Anne, Virginia A Marchman, and Adriana Weisleder. 2013. “SES Differences in Language Processing Skill and Vocabulary Are Evident at 18 Months.” *Developmental Science* 16 (2): 234–48.
- Fernald, Anne, Amy Perfors, and Virginia A. Marchman. 2006. “Picking up Speed in Understanding: Speech Processing Efficiency and Vocabulary Growth Across the 2nd Year.” *Developmental Psychology* 42 (1): 98–116. <https://doi.org/10.1037/0012-1649.42.1.98>.
- Fernald, Anne, John P. Pinto, Daniel Swingley, Amy Weinberg, and Gerald W. McRoberts. 1998. “Rapid Gains in Speed of Verbal Processing by Infants in the 2nd Year.” *Psychological Science* 9 (3): 228–31.
- Fernald, Anne, Renate Zangl, Ana Luz Portillo, and Virginia A. Marchman. 2008. “Looking While Listening: Using Eye Movements to Monitor Spoken Language Comprehension by Infants and Young Children.” In *Developmental Psycholinguistics: On-Line Methods in Children’s Language Processing*, edited by Irina A. Sekerina, Eva M. Fernandez, and Harald Clahsen, 97–135. Amsterdam: John Benjamins.
- Fourtassi, Abdellah, Yuan Bian, and Michael C Frank. 2020. “The Growth of Children’s Semantic and Phonological Networks: Insight from 10 Languages.” *Cognitive Science* 44 (7): e12847. <https://doi.org/10.1111/cogs.12847>.
- Frank, Michael C. 2023. “Bridging the Data Gap Between Children and Large Language Models.” *Trends in Cognitive Sciences* 27 (11): 990–92.
- Frank, Michael C, Mika Braginsky, Daniel Yurovsky, and Virginia A Marchman. 2021. *Variability and Consistency in Early Language Learning: The Wordbank Project*. MIT Press.
- Frank, Michael C, Mika Braginsky, Daniel Yurovsky, and Virginia A. Marchman. 2017. “Wordbank: An Open Repository for Developmental Vocabulary Data.” *Journal of Child Language* 44 (3): 677–94.
- Frank, Michael C, and Noah D Goodman. 2025. “Cognitive Modeling Using Artificial Intelligence.” *Annual Review of Psychology* 77.
- Frank, Michael C, Virginia A Marchman, Claire Augusta Bergey, Veronica Boyce, Mika Braginsky, George Kachergis, Jess Mankewitz, et al. 2026. “Continuous Developmental Changes in Word



- Recognition Support Language Learning Across Early Childhood.” *eLife* 14.
- Futrell, Richard, and Kyle Mahowald. 2025. “How Linguistics Learned to Stop Worrying and Love the Language Models.” *Behavioral and Brain Sciences*, 1–98.
- Ganger, Jennifer, and Michael R. Brent. 2004. “Reexamining the Vocabulary Spurt.” *Developmental Psychology* 40 (4): 621–32. <https://doi.org/10.1037/0012-1649.40.4.621>.
- Gómez Díaz, Miranda, Laia Fibla, Rachel Ka-Ying Tsui, and Krista Byers-Heinlein. 2024. “Testing Theories of the Vocabulary Spurt with Monolingual and Bilingual Infants.” *Developmental Psychology* 60 (8): 1357–71. <https://doi.org/10.1037/dev0001777>.
- Goodman, Judith C, Philip S Dale, and Ping Li. 2008. “Does Frequency Count? Parental Input and the Acquisition of Vocabulary.” *Journal of Child Language* 35 (3): 515–31.
- Grimm, Kevin J, Nilam Ram, and Ryne Estabrook. 2016. *Growth Modeling: Structural Equation and Multilevel Modeling Approaches*. Guilford Publications.
- Hagihara, Hiromichi, Monica Barbir, Mikako Ishibashi, Yasuhiro Kanakogi, Masaharu Kato, Irena Lovcevic, Youtao Lu, et al. 2023. “A Sharable Merged Dataset on Japanese Children’s Vocabulary Measures Using the Japanese MacArthur–Bates Communicative Development Inventory.” OSF. <https://doi.org/10.17605/osf.io/s5ydw>.
- Hart, Betty, and Todd R Risley. 1995. *Meaningful Differences in the Everyday Experience of Young American Children*. Brookes.
- . 2003. “The Early Catastrophe: The 30 Million Word Gap by Age 3.” *American Educator* 27 (1): 4–9.
- Hidaka, Shohei. 2013. “A Computational Model Associating Learning Process, Word Attributes, and Age of Acquisition.” *PloS One* 8 (11).
- Hills, Thomas T., Mounir Maouene, Josita Maouene, Adam Sheya, and Linda Smith. 2009. “Longitudinal Analysis of Early Semantic Networks: Preferential Attachment or Preferential Acquisition?” *Psychological Science* 20 (6): 729–39. <https://doi.org/10.1111/j.1467-9280.2009.02365.x>.
- Hirsh-Pasek, Kathy, Lauren B Adamson, Roger Bakeman, Margaret Tresch Owen, Roberta Michnick Golinkoff, Amy Pace, Paula KS Yust, and Katharine Suma. 2015. “The Contribution of Early Communication Quality to Low-Income Children’s Language Success.” *Psychological Science* 26 (7): 1071–83.
- Huebner, Philip A., Elior Sulem, Cynthia Fisher, and Dan Roth. 2021. “BabyBERTa: Learning More Grammar with Small-Scale Child-Directed Language.” In *Proceedings of the 25th Conference on Computational Natural Language Learning (CoNLL)*, 624–46. <https://doi.org/10.18653/v1/2021.conll-1.49>.
- Kachergis, George, Virginia A Marchman, Philip S Dale, Jessica Mankewitz, and Michael C Frank. 2022. “Online Computerized Adaptive Tests of Children’s Vocabulary Development in English and Mexican Spanish.” *Journal of Speech, Language, and Hearing Research* 65 (6): 2288–308.
- Kachergis, George, Virginia A Marchman, and Michael C Frank. 2022. “Toward a ‘Standard Model’ of Early Language Learning.” *Current Directions in Psychological Science* 31 (1): 20–27.
- Long, Bria, Robert Z Sparks, Violet Xiang, Stefan Stojanov, Zi Yin, Grace E Keene, Alvin WM Tan, et al. 2024. “The BabyView Dataset: High-Resolution Egocentric Videos of Infants’ and Young Children’s Everyday Experiences.” *arXiv Preprint arXiv:2406.10447*.
- MacWhinney, Brian. 2000. *The CHILDES Project: Tools for Analyzing Talk*. Vol. 1. Psychology Press.
- Marchman, Virginia A, Philip Dale, and Larry Fenson. 2021. *MacArthur-Bates Communicative*



- Development Inventories User's Guide and Technical Manual, Third Edition*. Brookes Publishing.
- Marchman, Virginia A, and Anne Fernald. 2008. "Speed of Word Recognition and Vocabulary Knowledge in Infancy Predict Cognitive and Language Outcomes in Later Childhood." *Developmental Science* 11 (3): F9–16.
- Markman, Ellen M., and Gwyn F. Wachtel. 1988. "Children's Use of Mutual Exclusivity to Constrain the Meanings of Words." *Cognitive Psychology* 20 (2): 121–57.
- Markson, Lori, and Paul Bloom. 1997. "Evidence Against a Dedicated System for Word Learning in Children." *Nature* 385 (6619): 813–15.
- Mayor, Julien, and Kim Plunkett. 2010. "Vocabulary Spurt: Are Infants Full of Zipf?" In *Proceedings of the 32nd Annual Conference of the Cognitive Science Society*. Vol. 32.
- McMurray, Bob. 2007. "Defusing the Childhood Vocabulary Explosion." *Science* 317 (5838): 631–31.
- Mitchell, C., and B. McMurray. 2009. "On Leveraged Learning in Lexical Acquisition and Its Relationship to Acceleration." *Cognitive Science* 33 (8): 1503–23.
- Mollica, Francis, and Steven T Piantadosi. 2017. "How Data Drive Early Word Learning: A Cross-Linguistic Waiting Time Analysis." *Open Mind* 1 (2): 67–77.
- Muter, Valerie, Charles Hulme, Margaret J Snowling, and Jim Stevenson. 2004. "Phonemes, Rimes, Vocabulary, and Grammatical Skills as Foundations of Early Reading Development: Evidence from a Longitudinal Study." *Developmental Psychology* 40 (5): 665.
- Peter, Michelle S, Samantha Durrant, Andrew Jessop, Amy Bidgood, Julian M Pine, and Caroline F Rowland. 2019. "Does Speed of Processing or Vocabulary Size Predict Later Language Growth in Toddlers?" *Cognitive Psychology* 115: 101238.
- Pinstock, Eveline, and Satyam Antonio Schramm. 2026. "Vocabulary Trajectories in German-Speaking Children from 18 Months to Three Years: A Growth Mixture Model." *Journal of Child Language* 53 (1): 62–79. <https://doi.org/10.1017/S0305000924000503>.
- Rasch, Georg. 1960. *Probabilistic Models for Some Intelligence and Attainment Tests*. Copenhagen, Denmark: Danish Institute for Educational Research.
- Rescorla, Leslie. 2011. "Late Talkers: Do Good Predictors of Outcome Exist?" *Developmental Disabilities Research Reviews* 17 (2): 141–50.
- Rowe, Meredith L. 2012. "A Longitudinal Investigation of the Role of Quantity and Quality of Child-Directed Speech in Vocabulary Development." *Child Development* 83 (5): 1762–74.
- Rowe, Meredith L, and Catherine E Snow. 2020. "Analyzing Input Quality Along Three Dimensions: Interactive, Linguistic, and Conceptual." *Journal of Child Language* 47 (1).
- Roy, Brandon C., Michael C Frank, Philip DeCamp, Matthew Miller, and Deb Roy. 2015. "Predicting the Birth of a Spoken Word." *Proceedings of the National Academy of Sciences* 112 (41): 12663–68.
- Sanchez, Alessandro, Stephan C. Meylan, Mika Braginsky, Kyle E. MacDonald, Daniel Yurovsky, and Michael C Frank. 2019. "Childe-Db: A Flexible and Reproducible Interface to the Child Language Data Exchange System." *Behavior Research Methods* 51 (4): 1928–41.
- Simonsen, Hanne Gram, Kristian Emil Kristoffersen, Dorte Bleses, Sonja Wehberg, and Rune Nørgård Jørgensen. 2014. "The Norwegian Communicative Development Inventories: Reliability, Main Developmental Trends and Gender Differences." *First Language* 34 (1): 3–23. <https://doi.org/10.1177/0142723713510997>.
- Singletary, Britt, Hui Jiang, Winifred Graham Wilberforce, Daniela Avelar, Kristina Strother-Garcia, and Laura M. Justice. 2025. "Examining Early Vocabulary Growth Trajectories in Late Talkers in a Low-Income Longitudinal Sample." *Infancy* 30 (4): e70036. <https://doi.org/10.1111/inf.70036>.

- Smith, Linda B., Susan S. Jones, Barbara Landau, Lisa Gershkoff-Stowe, and Larissa Samuelson. 2002. "Object Name Learning Provides on-the-Job Training for Attention." *Psychological Science* 13 (1): 13–19. <https://doi.org/10.1111/1467-9280.00403>.
- Sperry, Douglas E., Linda L. Sperry, and Peggy J. Miller. 2019. "Reexamining the Verbal Environments of Children from Different Socioeconomic Backgrounds." *Child Development* 90 (4): 1303–18. <https://doi.org/10.1111/cdev.13072>.
- Steyvers, Mark, and Joshua B. Tenenbaum. 2005. "The Large-Scale Structure of Semantic Networks: Statistical Analyses and a Model of Semantic Growth." *Cognitive Science* 29 (1): 41–78. [https://doi.org/10.1207/s15516709cog2901\\_3](https://doi.org/10.1207/s15516709cog2901_3).
- Thal, Donna J., Virginia A. Marchman, and J. Bruce Tomblin. 2013. "Late Talking Toddlers: Characterization and Prediction of Continued Delay." In *Late Talkers: Language Development, Interventions, and Outcomes*, edited by L. Rescorla and P. Dale. Baltimore, MD: Brookes Publishing.
- Trueswell, John C, Tamara Nicol Medina, Alon Hafri, and Lila R Gleitman. 2013. "Propose but Verify: Fast Mapping Meets Cross-Situational Word Learning." *Cognitive Psychology* 66 (1): 126–56.
- Warstadt, Alex, Aaron Mueller, Leshem Choshen, Ethan Wilcox, Chengxu Zhuang, Juan Ciro, Rafael Mosquera, et al. 2023. "Findings of the BabyLM Challenge: Sample-Efficient Pretraining on Developmentally Plausible Corpora." In *Proceedings of the BabyLM Challenge at the 27th Conference on Computational Natural Language Learning*, 1–34.
- Weisleder, A., and A. Fernald. 2013. "Talking to Children Matters: Early Language Experience Strengthens Processing and Builds Vocabulary." *Psychological Science* 24 (11): 2143–52. <https://doi.org/10.1177/0956797613488145>.
- Woodward, Amanda L, Ellen M Markman, and Colleen M Fitzsimmons. 1994. "Rapid Word Learning in 13-and 18-Month-Olds." *Developmental Psychology* 30 (4): 553.
- Zettersten, Martin, Daniel Yurovsky, Tian Linger Xu, Sarp Uner, Angeline Sin Mei Tsui, Rose M Schneider, Annissa N Saleh, et al. 2023. "Peekbank: An Open, Large-Scale Repository for Developmental Eye-Tracking Data of Children’s Word Recognition." *Behavior Research Methods* 55 (5): 2485–2500.

# Supplemental Information

## Comparison with Prior Models

## Relation to prior accumulator models [draft for promotion - MCF to review]

Our model is one member of a family of accumulator models of early vocabulary, and the family is easiest to compare in a single set of coordinates. Writing the latent ability of child  $i$  on word  $j$  at age  $t$  as

$$\eta_{ij}(t) = \log r_i + \log \alpha_i + \kappa_i \log t - \delta_j,$$

every prior model in this tradition is a special case that fixes one or more of three quantities: the between-child variance in efficiency ( $\sigma_\alpha$ ), the population scaling exponent ( $\kappa_{\text{pop}}$ ), and - critically - the between-child variance in that exponent ( $\sigma_\kappa$ ). Mapping each prior model into these coordinates makes the comparison structural rather than notational.

**McMurray [-@mcmurray2007]** is a pure accumulator: every word fills at a single unit rate ( $\kappa = 1$ ) and children do not differ ( $\sigma_\alpha = \sigma_\kappa = 0$ ). Its central insight is that the *vocabulary* curve accelerates even when the underlying rate does not, because ability sweeps through a distribution of word difficulties - the construction reproduced in panel (A) of our schematic. Acceleration here is a population-level property of the difficulty distribution, not a change in the child. This corresponds to our M1.

**Mitchell & McMurray [-@mitchell2009]** add a leverage term: already-known words speed the acquisition of new ones. They prove that this feedback shifts *when* words are acquired but cannot manufacture acceleration that is not already present in the difficulty distribution. In our coordinates it moves the timing of acquisition without changing  $\kappa$ , and - being non-linear in accumulated vocabulary - it is the one prior model that does not embed cleanly in the linear predictor above.

**Hidaka [-@hidaka2013]** introduces a rate-change process in which ability grows as  $t^{D+1}$ ; his exponent  $D$  is exactly  $\kappa_{\text{pop}} - 1$  in our notation, so  $D > 0$  is population-level acceleration ( $\kappa_{\text{pop}} > 1$ ). This is our M2. The acceleration is shared across children ( $\sigma_\kappa = 0$ ). Hidaka relates  $D$  to word imageability - i.e. a *per-word* scaling exponent - which is an interesting direction we do not pursue here.

**Mollica & Piantadosi [-@mollica2017]** model each word as a Gamma waiting-time process in which acquisition occurs after roughly ten "effective learning instances." In our coordinates this is again a pure accumulator ( $\sigma_{\alpha} = 1$ ); the per-word variation they capture lives in the Gamma shape parameter, which corresponds to a per-word discrimination (a 2PL extension) rather than to per-child structure.

**Kachergis et al. [-@kachergis2022]** are the first to free  $\sigma_{\alpha}$ : they cast vocabulary as a Rasch model in which each child has their own latent ability, unifying the accumulator and item-response traditions. This is our M3. But the scaling exponent remains shared across children ( $\sigma_{\kappa} = 0$ ), and the model is proposed as a framework rather than fit to longitudinal individual data.

Reading down this list, the literature progressively frees parameters - McMurray fixes all three, Hidaka frees  $\kappa_{\text{pop}}$ , Kachergis frees  $\sigma_{\alpha}$  - but every model holds  $\sigma_{\kappa} = 0$ .

| Model                       | $\sigma_{\alpha}$ | $\kappa_{\text{pop}}$ | $\sigma_{\kappa}$ | per-word $\delta_j$ |
|-----------------------------|-------------------|-----------------------|-------------------|---------------------|
| McMurray (2007)             | 0                 | 1                     | 0                 | free                |
| Mitchell & McMurray (2009)  | 0                 | 1                     | 0                 | free + leverage     |
| Hidaka (2013), rate-change  | 0                 | $D+1$                 | 0                 | free                |
| Mollica & Piantadosi (2017) | 0                 | 1                     | 0                 | free                |
| Kachergis et al. (2022)     | free              | free                  | 0                 | free                |
| <b>This work (M4)</b>       | <b>free</b>       | <b>free</b>           | <b>free</b>       | <b>free</b>         |

Our contribution is to free  $\sigma_{\kappa}$ : children differ not only in how efficient they are but in how steeply their learning accelerates. This is the load-bearing addition - it is what lets the model fit the widening fan of trajectories with age (M4 vs M3 in the main text). It is also a falsifiable claim rather than a parameter convenience: if  $\sigma_{\kappa}$  were near zero the model would collapse onto Kachergis et al., but the estimated per-child slope variance is well above zero and large relative to the efficiency variance  $\sigma_{\alpha}$  (main text, Figure 2 and Results).

## Nonidentifiability in Cross-Sectional Data

### Datasets Used

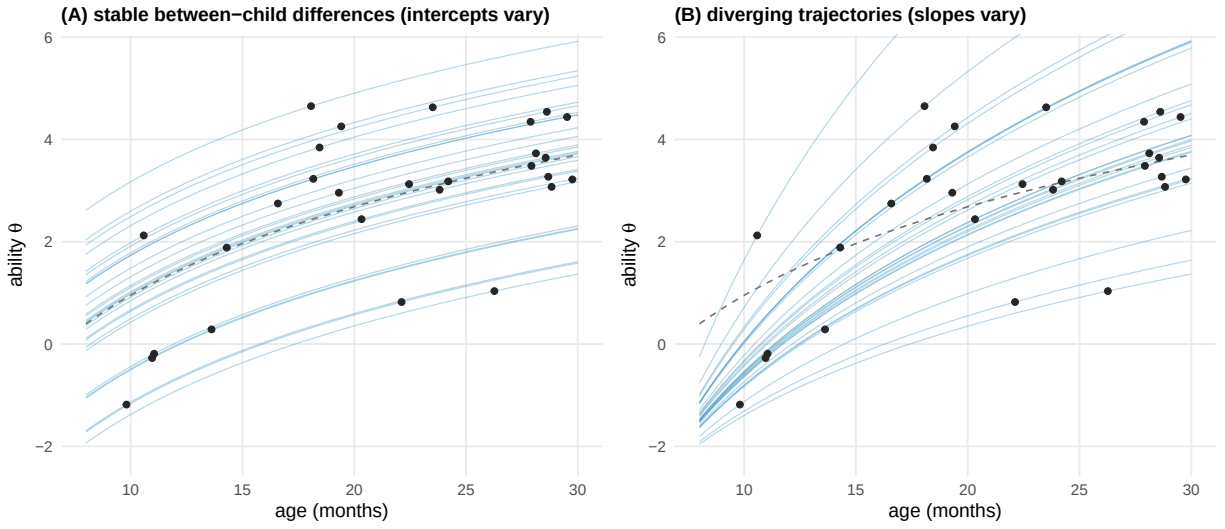


Figure 6: Non-identifiability of growth models in cross-sectional data. Panels show the same simulated data, reflecting the possibility of stable between-child intercept differences (A) vs. varying growth slopes (B).

Table 1: Characteristics of all datasets used in the paper. N and ages reflect the analysis samples (post-exclusion). Cross-sectional rows pool all contributed Wordbank datasets for a language. For AM2018 and FMW2013, which enter both the input and processing analyses with slightly different inclusion criteria, N is the union of the two subsamples.

| Citation                       | Language             | N     | Admins | Age (months) |     |     | Long. | Input   | Recordings | LWL sessions |
|--------------------------------|----------------------|-------|--------|--------------|-----|-----|-------|---------|------------|--------------|
|                                |                      |       |        | Mean         | Min | Max |       |         |            |              |
| Wordbank longitudinal          |                      |       |        |              |     |     |       |         |            |              |
| Thal et al. (20XX)             | English (American)   | 653   | 1,947  | 19.1         | 12  | 29  | yes   |         |            |              |
| Smith et al. (20XX)            | English (American)   | 314   | 878    | 21.9         | 16  | 30  | yes   |         |            |              |
| Marchman et al. (20XX)         | English (American)   | 314   | 714    | 22.6         | 16  | 30  | yes   |         |            |              |
| Simonsen et al. (2014)         | Norwegian            | 1,676 | 6,347  | 25.1         | 8   | 36  | yes   |         |            |              |
| Hagihara et al. (2023)         | Japanese             | 178   | 425    | 17.4         | 8   | 34  | yes   |         |            |              |
| Input / processing             |                      |       |        |              |     |     |       |         |            |              |
| Long et al. (2024)             | English (American)   | 22    | 103    | 18.6         | 9   | 32  | yes   | headcam | 5,739      |              |
| Egan-Dailey & Bergelson (2025) | English (American)   | 44    | 514    | 12.2         | 6   | 18  | yes   | LENA    | 525        |              |
| Adams et al. (2018)            | English (American)   | 67    | 167    | 20.1         | 13  | 27  | yes   | LENA    | 126        | 440          |
| Fernald et al. (2013)          | English (American)   | 60    | 142    | 23.2         | 18  | 30  | yes   | LENA    | 51         | 93           |
| Fernald & Marchman (2012)      | English (American)   | 119   | 393    | 22.9         | 14  | 30  | yes   |         |            | 419          |
| Wordbank cross-sectional       |                      |       |        |              |     |     |       |         |            |              |
| Frank et al. (2017)            | Arabic (Saudi)       | 258   | 258    | 11.9         | 8   | 16  |       |         |            |              |
| Frank et al. (2017)            | Cantonese            | 1,208 | 1,208  | 23.4         | 16  | 30  |       |         |            |              |
| Frank et al. (2017)            | Catalan              | 1,436 | 1,436  | 19.7         | 8   | 30  |       |         |            |              |
| Frank et al. (2017)            | Croatian             | 627   | 627    | 19.0         | 8   | 30  |       |         |            |              |
| Frank et al. (2017)            | Czech                | 493   | 493    | 23.0         | 16  | 30  |       |         |            |              |
| Frank et al. (2017)            | Danish               | 6,112 | 6,112  | 20.8         | 8   | 36  |       |         |            |              |
| Frank et al. (2017)            | Dutch                | 446   | 446    | 15.6         | 8   | 31  |       |         |            |              |
| Frank et al. (2017)            | English (American)   | 8,685 | 8,685  | 19.2         | 8   | 30  |       |         |            |              |
| Frank et al. (2017)            | English (Australian) | 1,493 | 1,493  | 20.6         | 12  | 30  |       |         |            |              |
| Frank et al. (2017)            | Estonian             | 1,235 | 1,235  | 23.3         | 16  | 30  |       |         |            |              |
| Frank et al. (2017)            | Finnish              | 234   | 234    | 16.5         | 9   | 24  |       |         |            |              |
| Frank et al. (2017)            | French (French)      | 1,290 | 1,290  | 21.6         | 8   | 30  |       |         |            |              |
| Frank et al. (2017)            | German               | 1,181 | 1,181  | 24.1         | 18  | 30  |       |         |            |              |
| Frank et al. (2017)            | Hebrew               | 534   | 534    | 27.4         | 11  | 35  |       |         |            |              |
| Frank et al. (2017)            | Hungarian            | 369   | 369    | 23.1         | 16  | 30  |       |         |            |              |
| Frank et al. (2017)            | Italian              | 1,400 | 1,400  | 20.4         | 7   | 36  |       |         |            |              |
| Frank et al. (2017)            | Japanese             | 846   | 846    | 20.4         | 8   | 36  |       |         |            |              |
| Frank et al. (2017)            | Kigiriama            | 205   | 205    | 16.1         | 8   | 30  |       |         |            |              |
| Frank et al. (2017)            | Korean               | 2,229 | 2,229  | 22.9         | 8   | 36  |       |         |            |              |
| Frank et al. (2017)            | Latvian              | 683   | 683    | 22.3         | 8   | 36  |       |         |            |              |
| Frank et al. (2017)            | Mandarin (Beijing)   | 1,056 | 1,056  | 23.0         | 16  | 30  |       |         |            |              |
| Frank et al. (2017)            | Mandarin (Taiwanese) | 2,654 | 2,654  | 22.0         | 8   | 36  |       |         |            |              |
| Frank et al. (2017)            | Norwegian            | 7,358 | 7,358  | 23.3         | 8   | 36  |       |         |            |              |

(continued)

| Citation            | Language              | N     | Admins | Age (months) |     |     | Long. | Input | Recordings | LWL sessions |
|---------------------|-----------------------|-------|--------|--------------|-----|-----|-------|-------|------------|--------------|
|                     |                       |       |        | Mean         | Min | Max |       |       |            |              |
| Frank et al. (2017) | Portuguese (European) | 4,322 | 4,322  | 19.7         | 8   | 30  |       |       |            |              |
| Frank et al. (2017) | Russian               | 1,803 | 1,803  | 21.4         | 8   | 36  |       |       |            |              |
| Frank et al. (2017) | Slovak                | 1,712 | 1,712  | 20.0         | 8   | 36  |       |       |            |              |
| Frank et al. (2017) | Spanish (Argentinian) | 783   | 783    | 23.3         | 16  | 33  |       |       |            |              |
| Frank et al. (2017) | Spanish (European)    | 1,005 | 1,005  | 18.4         | 8   | 30  |       |       |            |              |
| Frank et al. (2017) | Spanish (Mexican)     | 1,915 | 1,915  | 19.1         | 8   | 30  |       |       |            |              |
| Frank et al. (2017) | Swedish               | 1,358 | 1,358  | 19.0         | 8   | 28  |       |       |            |              |
| Frank et al. (2017) | Turkish               | 3,537 | 3,537  | 21.4         | 8   | 36  |       |       |            |              |

## AIC and BIC for Longitudinal Models

Table 2

| Model                   | Age    | $\Delta$ AIC |         |          |           |          |
|-------------------------|--------|--------------|---------|----------|-----------|----------|
|                         |        | Thal         | Smith   | Marchman | Norwegian | Japanese |
| M1. accumulator         | —      | 703,220      | 273,732 | 251,211  | 2,281,606 | 94,503   |
| M2. + acceleration      | linear | 206,553      | 179,872 | 134,596  | 1,377,615 | 41,169   |
| M2. + acceleration      | log    | 204,930      | 178,471 | 133,877  | 1,348,013 | 39,054   |
| M3. + efficiency var.   | linear | 35,856       | 17,418  | 13,752   | 130,625   | 3,850    |
| M3. + efficiency var.   | log    | 34,219       | 15,356  | 13,135   | 115,116   | 3,227    |
| M4. + acceleration var. | linear | 1,134        | 553     | 0        | 4,301     | 169      |
| M4. + acceleration var. | log    | 0            | 0       | 30       | 0         | 0        |

Table 3

| Model                   | Age    | $\Delta$ BIC |         |          |           |          |
|-------------------------|--------|--------------|---------|----------|-----------|----------|
|                         |        | Thal         | Smith   | Marchman | Norwegian | Japanese |
| M1. accumulator         | —      | 703,172      | 273,687 | 251,166  | 2,281,553 | 94,461   |
| M2. + acceleration      | linear | 206,517      | 179,838 | 134,563  | 1,377,575 | 41,138   |
| M2. + acceleration      | log    | 204,895      | 178,437 | 133,843  | 1,347,973 | 39,023   |
| M3. + efficiency var.   | linear | 35,832       | 17,396  | 13,729   | 130,599   | 3,829    |
| M3. + efficiency var.   | log    | 34,195       | 15,333  | 13,113   | 115,089   | 3,206    |
| M4. + acceleration var. | linear | 1,134        | 553     | 0        | 4,301     | 169      |
| M4. + acceleration var. | log    | 0            | 0       | 30       | 0         | 0        |

## Full Cross-Linguistic Model Ladder

## Population Input Range Estimation

Table 4

| Source                                  | Sample                | Channel          | n  | tok/hr | $\mu$ SD  | tok/mo | $\sigma_r$ |
|-----------------------------------------|-----------------------|------------------|----|--------|-----------|--------|------------|
| Hart & Risley                           | all kids pooled       | mother CDS       | 42 | 1,148  | 546–2,410 | 419K   | 0.74       |
| Hart & Risley                           | within-stratum pooled | mother CDS       | 42 | 1,148  | 631–2,086 | 419K   | 0.60       |
| Soderstrom & Wittebolle                 | all kids pooled       | all speech       | 12 | 967    | 622–1,502 | 353K   | 0.44       |
| Sperry                                  | all kids pooled       | adult-to-child   | 42 | 1,537  | 901–2,622 | 561K   | 0.53       |
| Sperry                                  | within-stratum pooled | adult-to-child   | 42 | 1,537  | 974–2,425 | 561K   | 0.46       |
| Weisleder & Fernald                     | all kids pooled       | mother CDS       | 29 | 458    | 197–1,065 | 167K   | 0.84       |
| Bergelson 2018 (SEEDLingS, LENA)        | US daylong, 6–7 mo    | all adult (AWC)  | 44 | 1,101  | 767–1,580 | 402K   | 0.36       |
| Casillas 2020 (Tseltal Mayan)           | Chiapas daylong       | child-directed   | 10 | 720    | —         | 263K   | —          |
| Coffey et al. 2024 (plausible envelope) | cross-cultural bounds | CDS to all-input | —  | —      | 4–7,232   | —      | —          |

## Estimating the input-rate prior: units and conversions [draft for promotion - MCF to review]

The model represents each child's input as a log rate,  
 $\log r_i \sim \text{cal}(\mu_r, \sigma_r^2)$ , measured in **adult tokens per waking hour**. @tbl-input-rates collects published estimates of this



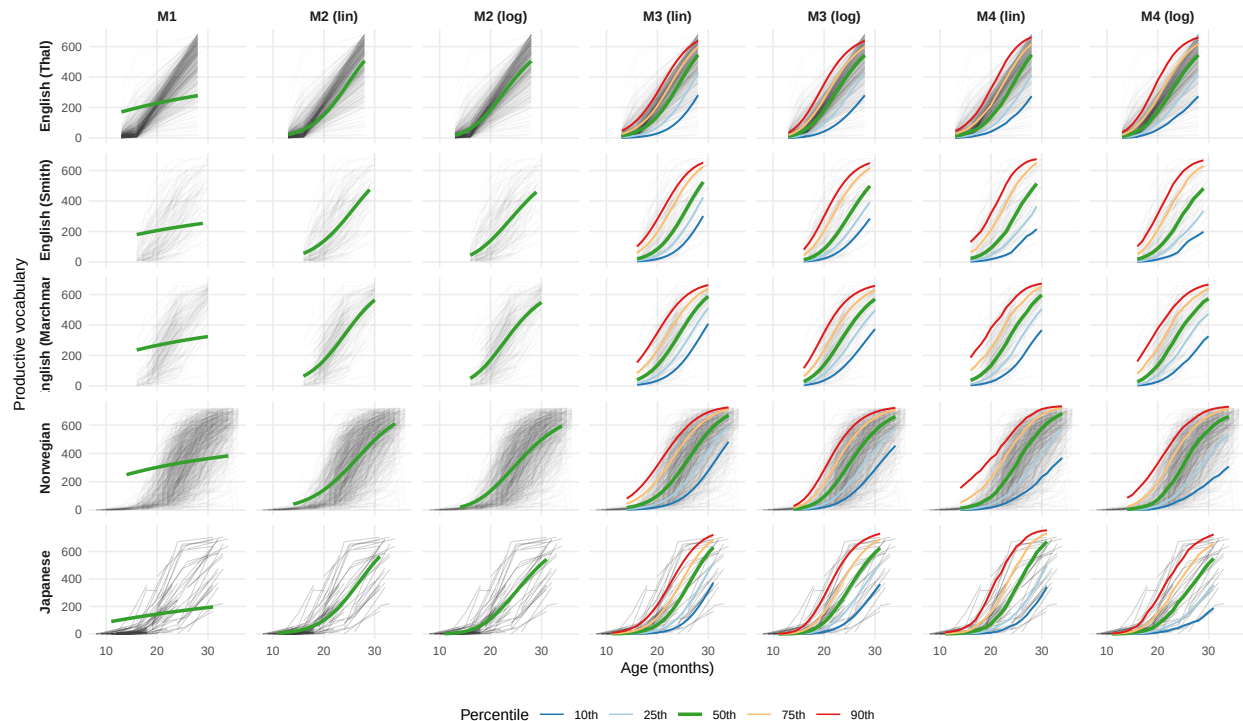


Figure 7

rate, expressed in shared units so they can be compared and so that the implied between-child spread  $\sigma_r$  is visible on one scale.

Conventions used throughout:

- **Tokens/hour to tokens/month.** We multiply by  $H = 365$  waking hours per month ( $\text{hr/day} \times 30.44 \text{ days/mo}$ ), the model's constant.
- **$\sigma_r$  is on the log scale** - the SD of  $\log(\text{tokens/hr})$  across children within a sample. Its exponential  $e^{\sigma_r}$  is the multiplicative factor for a one-SD shift in input (e.g.  $\sigma_r = 0.53 \rightarrow 1.7 \times$ ), and the 16th-84th percentile spread is  $e^{2\sigma_r}$ .
- **Channel.** Studies count different speaker sets; we take the channel closest to "tokens reaching the child" - adult-to-child for Sperry, all-adult LENA word counts for SEEDLingS, and maternal CDS for Hart & Risley and Weisleder & Fernald.
- **Pooled vs. within-stratum.** For Hart & Risley and Sperry we report  $\sigma_r$  both across all children (which mixes in SES and site gradients) and within stratum (subtracting each band's mean first); the latter is the cleaner within-population estimate.
- **Minutes to tokens.** Where a source reports minutes of speech per hour

rather than token counts (Casillas; the Coffey bounds), we convert at  $\approx 200$  words/min ( $0.3$  s/word, the Brent-corpus CDS rate used by @coffey2024); those rows are accurate to roughly  $\pm 20\%$ .

Rows marked *\*computed\** are recomputed here from per-dyad data; *\*literature\** rows are taken from the curated validation set in ``input_estimation/``.

## $\sigma_r$ Sensitivity Analysis

## Analytic input-rate ( $\sigma_r$ ) sensitivity [draft for promotion - MCF to review]

The input/efficiency variance split is identified only by the external pin on the population SD of log input rate,  $\sigma_r$  - we do not observe input in the longitudinal data. Because every reported input share therefore rests on a quantity we assume rather than estimate, we map how the conclusion moves as  $\sigma_r$  ranges over its plausible values (panel B of @fig-io-partition).

The data identify the *\*total\** between-child intercept variance,  $\sigma_{\xi}^2 = \sigma_{\alpha}^2 + \sigma_r^2$  (the spread of children's ability at the reference age  $a_0$ ), very precisely; the pin only decides how that fixed total is *\*split\** into an efficiency part  $\sigma_{\alpha}^2$  and an input part  $\sigma_r^2$ . Holding  $\sigma_{\xi}^2$  at its fitted value, the input share is then a closed form,

$$1 - \pi_{\alpha}(\sigma_r) = \frac{\sigma_r^2}{\sigma_{\xi}^2},$$

which we evaluate across the plausible  $\sigma_r$  range *\*without refitting\** - the analytic sensitivity curve.

This is first-order: it treats  $\sigma_{\xi}^2$  as itself invariant to the pin. Refits at off-pin values reveal a small second-order feedback - pinning a larger  $\sigma_r$  slightly *\*shrinks\** the estimated total variance (for English,  $\sigma_{\xi}^2$  falls from 3.57 at  $\sigma_r=0.53$  to 3.22 at  $\sigma_r=0.80$ ). The analytic curve therefore slightly *\*understates\** the input share at large  $\sigma_r$  (18% analytic vs. 20% by refit at  $\sigma_r=0.80$ ), an error of about two points of  $\pi_{\alpha}$  in the conservative direction for the size of the input effect. The open points in panel B are these refit anchors; the gap between point and curve is the second-order term.

We use the analytic curve, anchored by the few refits we have, rather than a dense refit sweep: the qualitative conclusion - input accounts for a modest share even at the top of the plausible  $\sigma_r$  range - is unaffected by the  $\sim 2$ -point correction, whereas a full sweep of the Norwegian model alone would cost roughly half a day of compute per pin value.

## Comparing children and language models on one slope axis [draft for promotion - MCF to rev]

The comparison to language models in Figure 6 places children and models on a single, literally shared axis: the slope of the acquisition sigmoid in log-experience. Both populations have sigmoidal per-word acquisition curves, and the slope of that sigmoid is directly comparable.

On the child side, our linear predictor makes this slope exact. Differentiating  $\eta_{ij}(t)$  with respect to  $\log t$  gives  $\kappa_i$ : the logit-probability that a child produces a word rises linearly in log-age with slope equal to that child's scaling exponent. The per-child sigmoid slope *is*  $\kappa_i$ , by construction.

On the model side, @chang2022 fit a four-parameter logistic to each (model, word) pair, describing how the model's surprisal for a word falls over training. Re-expressed as the fraction of a word's surprisal reduction achieved so far, their fit is a sigmoid in  $\log(\text{training steps})$  with slope  $1/\text{scale}_w$ . Dividing by  $\ln 10$  converts this to natural-log units - the same units as  $\kappa_i$  - so the two slopes can be read off the same axis.

The result is a striking separation (Figure 6). Language models cluster near a slope of one - the unit-accumulator boundary, where each log-unit of additional experience yields a fixed increment of learning. Children sit in a broad distribution roughly an order of magnitude higher. On the shared axis, children's word learning accelerates with experience in a way that these models' does not.

Three caveats bound this comparison. First, the variability is asymmetric: children's slopes vary across *children*, whereas a single model is a deterministic learner, so model slopes vary across *words* within a run rather than across replicate learners. Second, the steep child slope is the *per-child* quantity; a researcher fitting a single sigmoid to cross-sectional Wordbank data per word would recover a shallower slope, because marginalizing over the population attenuates it. Third, the models in @chang2022 were trained on adult written text rather than child-directed speech, so part of the gap could reflect the input distribution rather than the learner. Models trained directly on CHILDES (Feng et al., forthcoming) provide the matched-input test: if the gap largely persists, it is structural; if it shrinks substantially, the input distribution is doing more of the work than we currently attribute to it.

<!-- [BIB NEEDED] Feng et al. (CHILDES-trained GPT) - no verified entry yet; named in text only. Daunizeau (2017) logistic-Gaussian approximation is used in the derivation note but not needed for this prose. -->

## Connection to neural scaling laws [draft for promotion - MCF to review; most speculative of

The per-word picture above connects to a more familiar aggregate phenomenon: neural scaling laws. A model's loss falls as a smooth power law in the amount of training data, and this descent is remarkably regular - different runs, and different model sizes, trace nearly the same curve (Kaplan et al., 2020; Hoffmann et al., 2022). Our per-word analysis is the micro-level counterpart of that regularity. Each word is acquired along a sigmoid whose slope sits near the unit-accumulator value, and the spread of those slopes - across words and across runs - is small. The aggregate smoothness of scaling laws and the per-word  $\kappa \approx 1$  we recover are two views of the same property: language models convert experience into knowledge as near-pure accumulators, with little instance-to-instance variation.

Children depart from this picture on both axes at once. Their per-instance scaling is far steeper ( $\kappa \approx 10$ ), and they vary substantially from one another in that exponent ( $\sigma_\kappa$  well above zero). The "data gap" between children and language models [Frank2023; Warstadt2023] is therefore not only that children reach adult competence from orders of magnitude less input. It is that children convert experience into knowledge through a different scaling relationship - steeper, and far more variable across individuals - than the one that scaling laws describe for models.

This connection is analogical rather than formal: scaling laws characterize aggregate loss, whereas our  $\kappa$  characterizes per-item acquisition, and the two need not move together. We offer it as a framing for why the data gap may be harder to close than raw data counts suggest.

<!-- [BIB NEEDED] Kaplan et al. (2020) and Hoffmann et al. (2022) are named in text but have NO verified bib entries yet - do not promote with live @ markers until added. -->

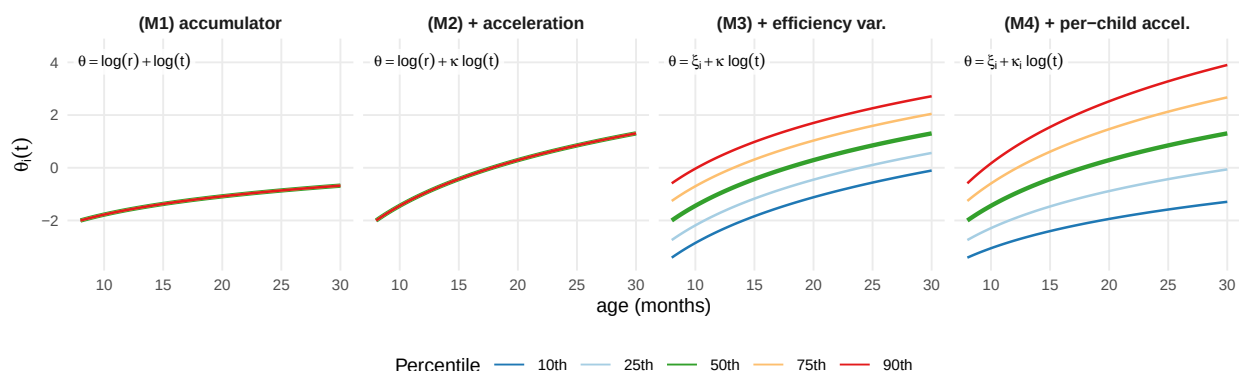


Figure 8

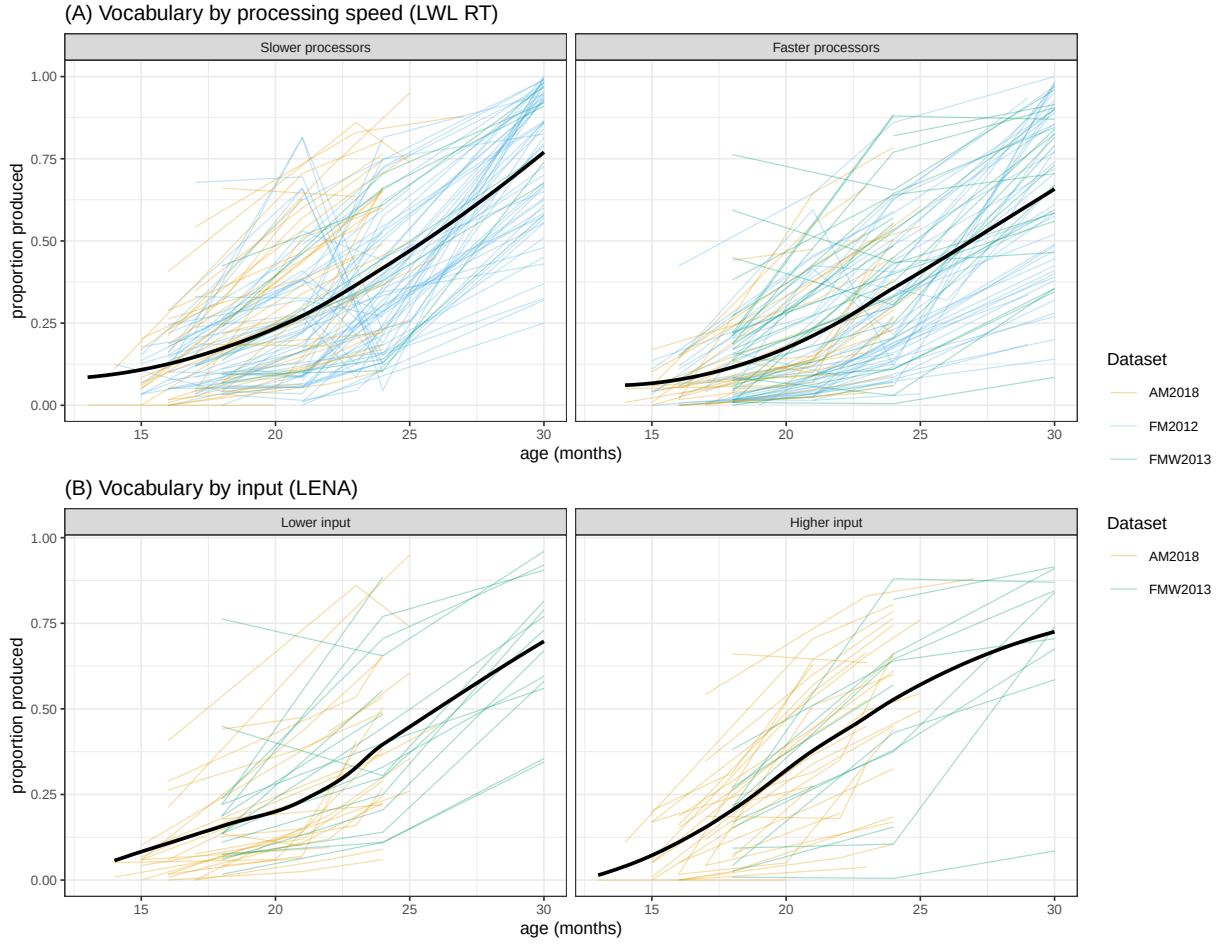


Figure 9

Table 5

| Dataset            | N    | Efficiency ( $\xi$ ) | Acceleration ( $\zeta$ ) |
|--------------------|------|----------------------|--------------------------|
| English (Thal)     | 651  | 0.96                 | >0.99                    |
| English (Smith)    | 312  | 0.30                 | 0.99                     |
| English (Marchman) | 313  | 0.99                 | 0.99                     |
| Norwegian          | 1667 | >0.99                | 0.86                     |
| Japanese           | 176  | 0.85                 | 0.99                     |
| Pooled             | 3119 | 0.98                 | 0.96                     |

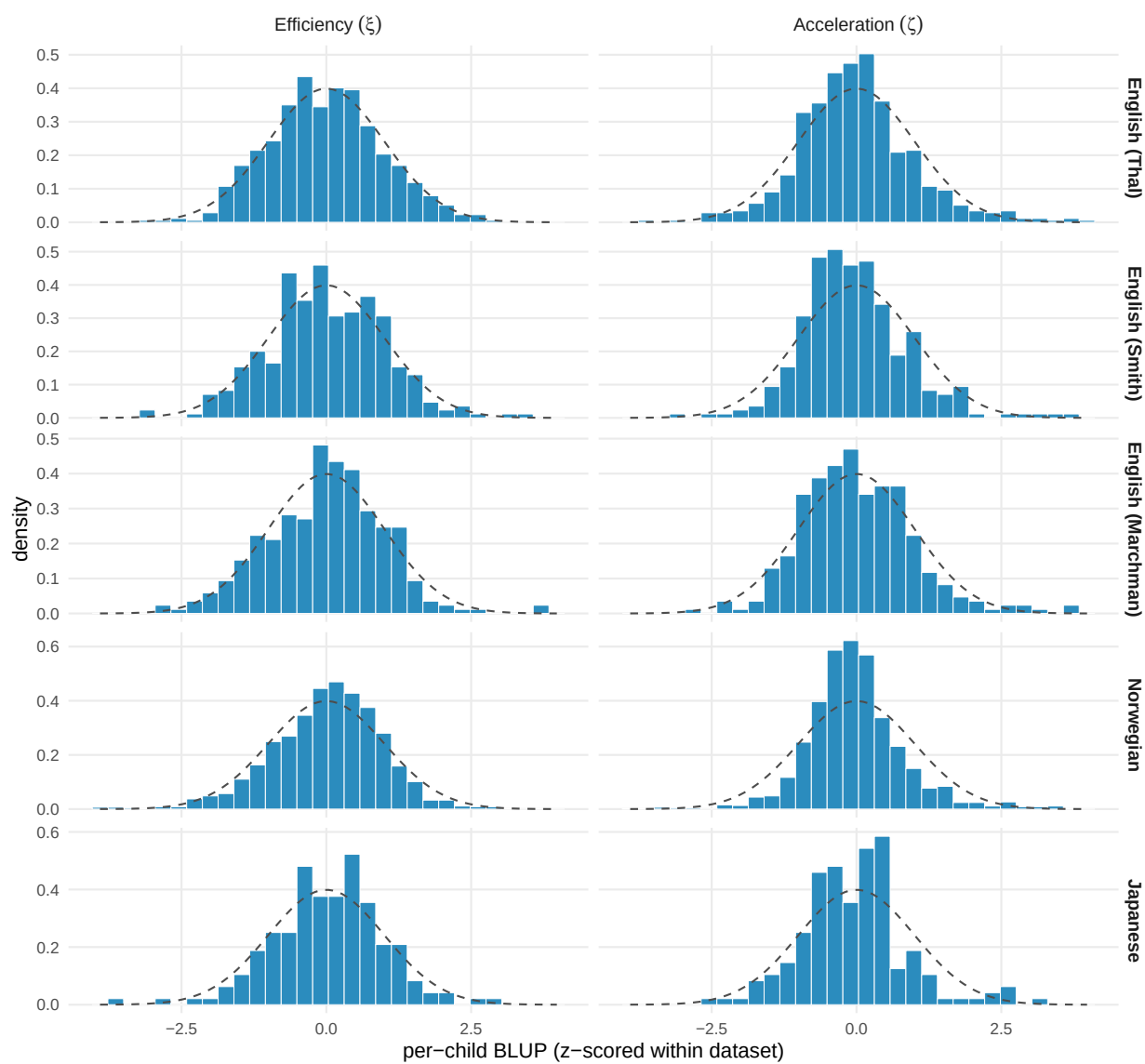


Figure 10